Detection of the hepatotoxic microcystins in 36 kinds of cyanobacteria Spirulina food products in China.

Gel filtration chromatography, ultra-filtration, and solid-phase extraction silica gel clean-up were evaluated for their ability to remove microcystins selectively from extracts of cyanobacteria Spirulina samples after using the reversed-phase octadecylsilyl ODS cartridge for subsequent analysis by liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS). The reversed-phase ODS cartridge/silica gel combination were effective and the optimal wash and elution conditions were: H(2)O (wash), 20% methanol in water (wash), and 90% methanol in water (elution) for the reversed-phase ODS cartridge, followed by 80% methanol in water elution in the silica gel cartridge. The presence of microcystins in 36 kinds of cyanobacteria Spirulina health food samples obtained from various retail outlets in China were detected by LC-MS/MS, and 34 samples (94%) contained microcystins ranging from 2 to 163 ng g(-1) (mean = 14 +/- 27 ng g(-1)), which were significantly lower than microcystins present in blue green alga products previously reported. MC-RR - which contains two molecules of arginine (R) - (in 94.4% samples) was the predominant microcystin, followed by MC-LR - where L is leucine - (30.6%) and MC-YR - where Y is tyrose - (27.8%). The possible potential health risks from chronic exposure to microcystins from contaminated cyanobacteria Spirulina health food should not be ignored, even if the toxin concentrations were low. The method presented herein is proposed to detect microcystins present in commercial cyanobacteria Spirulina samples.